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NATIONAL DEVELOPMENT COMPLEX
Deputy Directorate General of Quality Control
Physical Laboratory: Site-3A
TEST REPORT

Test Report No: TR (074)/PL/2022

Job Card No: JC (122)/PL/2022

Date: 07-07-2022

INDENTER'S DATA			
Name:	Usama Masood	Designation:	HOD (Preform)
Group & Site:	Preform / 3T	Directorate/ Department:	3DC-P / CPS
Organization:	NDC	Contact No:	051-9314439-41, Ext. 216

GENERAL DATA			
Sample Name:	3D Carbon Composite	ID No:	RD-20C/CTE-01~03
Lot No :	3D/CCC/Jun 21 / L-09	Lot Size:	----
Shape:	Cylindrical	Chemical Composition:	Carbon
Project:	S1AM, S2, S3, NASR, AWS, Herbah	Quantity:	03
Sample Receiving Date:	06-07-2022	Test Date:	06-07-2022

TEST DATA			
Test Method Used	PL/SOP/07	Qualification Criteria	NDC- CPS/Comp/QM/3DCarbon/CCC- 01, Rev : 01
Instrument Type	DIL-402C	Testing Temperature(°C)	RT~100
Temperature(°C)	25.5 ±1	Relative Humidity (%)	36 ±2
Test Type	Sample ID	Test Results	
Coefficient of Linear Thermal Expansion	RD-20C/CTE-01	-1.6649 E-06 1/K at 40°C	
		0.2027 E-06 1/K at 100°C	
	RD-20C/CTE-02	-0.4442 E-06 1/K at 40°C	
		0.7012 E-06 1/K at 100°C	
	RD-20C/CTE-03	-2.2545 E-06 1/K at 40°C	
		-0.3175 E-06 1/K at 100°C	
Test Status	Qualified for above mentioned test i.e. $\leq 2.0 \text{ E-06 1/K}$		
Remarks	i) These results apply to the items tested as received and mentioned above. ii) This test report shall not be reproduced except in full without the written approval of the laboratory. iii) The graphs of the samples analyzed are attached herewith. iv) Expanded Uncertainty is ± 0.000678 at 400°C at 95% confidence level & k=2.		

Test Performed By:	Imran Shahzad ARO	Contact No.	3301
Verified By:	Haseeb Ahmad Khan AM(T)	Contact No.	3301
Reviewed By:	Amir Khan HoD (CA)	Contact No.	3301
Approved By:	Eng. Dr. Nabi Bakhsh PD (ME)	Contact No.	3141